

Experiment 3

Module Name: Automation With Selenium

Experiment Title: Dynamic Dropdowns & Checkboxes

Problem Statement: Automate interactions with multiple checkboxes and a dynamic, auto-suggest dropdown menu.

Objective: To verify element states (`is_selected()`), send keystrokes to dynamic inputs, and iterate through lists of elements to find a match.

Tools, software, and concepts used: Python, Selenium, `find_elements()`, Python for loops.

Source code or implementation steps:

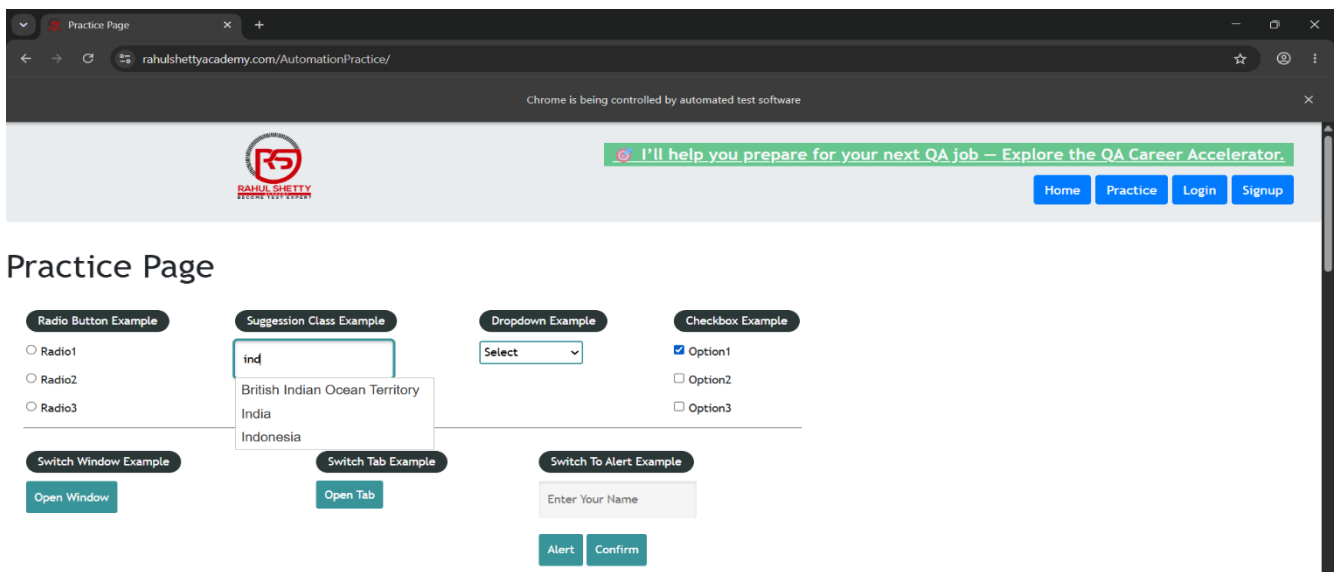
```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
driver = webdriver.Chrome()
driver.maximize_window()
try:
    driver.get("https://rahulshettyacademy.com/AutomationPractice/")

    # Checkbox Interaction
    checkbox1 = driver.find_element(By.ID, "checkBoxOption1")
    checkbox1.click()
    assert checkbox1.is_selected(), "Checkbox 1 was not selected."
    print("Checkbox successfully selected and verified.")

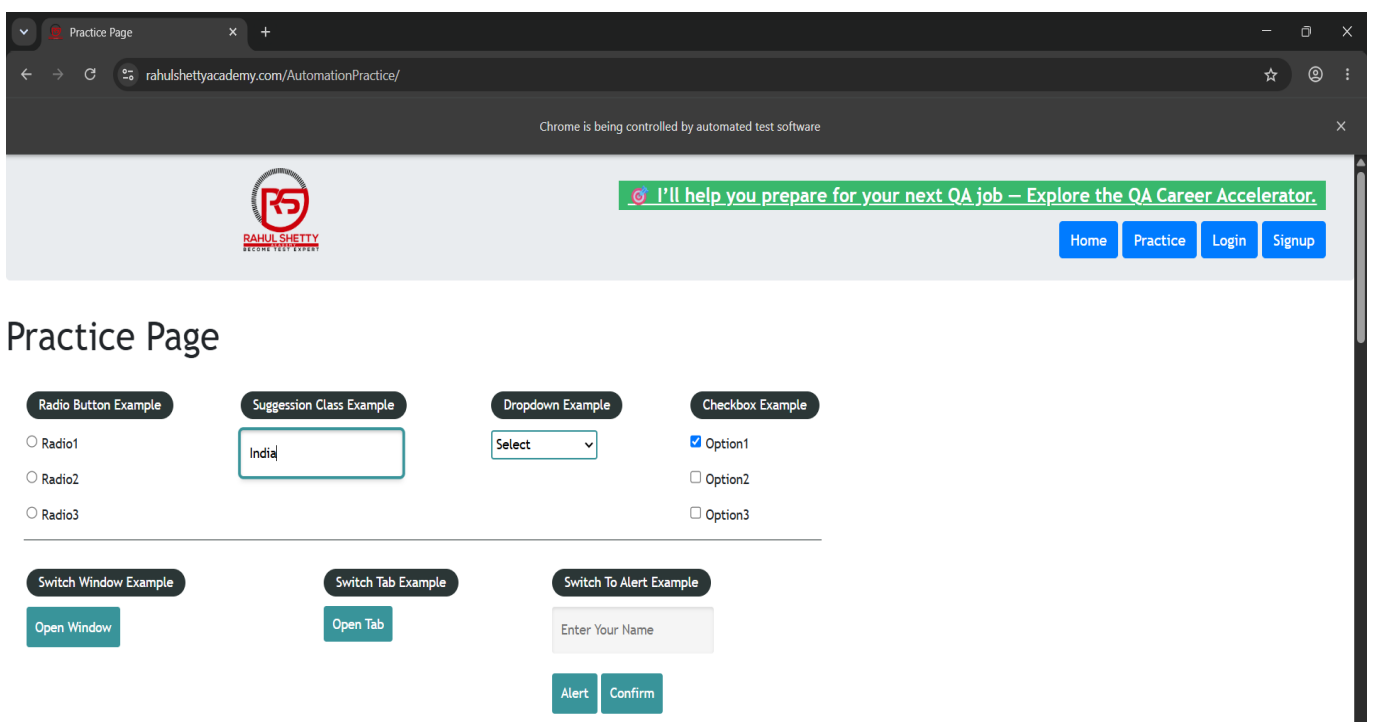
    # Dynamic Dropdown Interaction
    dropdown_input = driver.find_element(By.ID, "autocomplete")
    dropdown_input.send_keys("ind")
    time.sleep(2) # Wait for suggestions to populate

    # Capture all suggestions and loop through them
    suggestions = driver.find_elements(By.CSS_SELECTOR, "li.ui-menu-item div")
    for option in suggestions:
        if option.text == "India":
            option.click()
            break
    # Verify input value
    assert dropdown_input.get_attribute("value") == "India"
    print("Dynamic dropdown option successfully selected.")
finally:
    driver.quit()
```

Output with clearly labelled screenshots:



Screenshot 1: Checkbox 1 checked and Dropdown showing



Screenshot 2: "India" populated in the text box]

Console Output: Checkbox successfully selected and verified. \n Dynamic dropdown option successfully selected.

Brief result: Correctly checked boxes and parsed a dynamic list to click a specific text match.

Observation: find_elements() returns a list. Iterating through this list and extracting .text is a robust way to handle dynamic dropdowns where absolute XPaths might fail.

Conclusion: Logic and looping are required for complex, modern UI controls.